

Mathematics Extension 2

HSC Marking Feedback 2019

Question 11 **IMPORTANT:**

This resource was developed to support a previous version of the syllabus.

Part (a) (i) It may contain content that differs from the current syllabus.

Students should:

- be able to carry out complex number arithmetic.

In better responses, students were able to:

- use basic complex number arithmetic to write down the answer.

Areas for students to improve include:

- finding the correct complex conjugate.

Part (a) (ii)

Students should:

- be able to carry out complex number arithmetic.

In better responses, students were able to:

- multiply the numerator and denominator of $\frac{z}{w}$ by $\bar{w}$, the conjugate of the denominator.

Areas for students to improve include:

- being familiar with basic complex number products such as $(x + iy)(x - iy) = x^2 + y^2$.

Part (b)

Students should:

- be able sketch a conic by finding the eccentricity, foci and directrices.

In better responses, students were able to:

- provide a clear sketch showing all required features: intercepts, foci and directrices.

Areas for students to improve include:

- showing all necessary calculations used in finding the eccentricity of an ellipse and its key features
- using at least a third of a page for a graph or sketch.

Part (c)

Students should:

- be able to carry out integration of an expression involving a quadratic denominator.

In better responses, students were able to:

- write the denominator of the integrand in completed square form and apply a standard integral formula.

Areas for students to improve include:

- recognising the form of an integral and the most appropriate technique to apply to evaluate it
- making use of formulae given in examinations, such as the Reference Sheet results
- remembering to include an arbitrary constant ($+c$) in the primitive.

Part (d)

Students should:

- be able to carry out integration of an expression involving a quadratic denominator, using either partial fractions or an appropriate substitution.

In better responses, students were able to:

- write the integrand as a sum of partial fractions before integrating to complete the solution.

Areas for students to improve include:

- using a substitution only if it simplifies an integral rather than complicates it.

Part (e) (i)

Students should:

- be able to carry write a complex number in modulus-argument form.

In better responses, students were able to:

- evaluate the modulus and argument of z and proceed to write z in modulus-argument form.

Areas for students to improve include:

- identifying the correct quadrant in which a complex number lies in order to evaluate the correct argument.

Part (e) (ii)

Students should:

- be able to apply de Moivre's theorem and change from modulus-argument form to Cartesian form.

In better responses, students were able to:

- apply de Moivre's theorem to find z^3 in modulus-argument form before writing it in Cartesian form.

Areas for students to improve include:

- choosing the most efficient way to evaluate the power of a complex number, that is by applying de Moivre's theorem or by algebraic expansion.

Question 12

Part (a)

Students should:

- draw a clear and accurate sketch labelling all relevant detail.

In better responses, students were able to:

- graph $\arg(z) = \frac{\pi}{4}$ including an open circle at (0,0) and with $\frac{\pi}{4}$ labelled
- graph the line $y = 1$ with a label on the y -axis
- shade the correct region from the inequality with solid lines.

Areas for students to improve include:

- including an open circle when sketching $\arg z = A$
- labelling all critical information.

Part (b)(i)

Students should:

- show the appropriate substitution into the cosine rule.

In better responses, students were able to:

- substitute the angle $\frac{2\pi}{3}$ into the cosine rule then simplify the equation to gain the given result.

Areas for students to improve include:

- clearly showing all steps when the result is given.

Part (b)(ii)

Students should:

- apply the rules of implicit differentiation to find an expression for $\frac{dx}{dt}$
- substitute given values into their result.

In better responses, students were able to:

- interpret given information to obtain $\frac{dy}{dt} = 4$
- implicitly differentiate with respect to t to gain an expression involving $x, y, \frac{dx}{dt}$ and $\frac{dy}{dt}$ then

substitute to gain $\frac{dx}{dt}$

- implicitly differentiate with respect to x to gain expression involving x, y and $\frac{dy}{dx}$ then use the chain rule to obtain $\frac{dx}{dt}$ and substitute to find correct value.

Areas for students to improve include:

- understanding that the distance is measured from P to B so if the train is moving away from P the velocity is positive giving $\frac{dy}{dt} = 4$
- including $\frac{dy}{dt}$ when differentiating y with respect to t
- using the product rule in implicit differentiation.

Part (c)

Students should:

- interpret the information given to develop an integral then integrate to find the volume.

In better responses, students were able to:

- sketch a shell with length, height and depth and develop an expression for ΔV
- develop a volume expression that includes limits
- use integration by parts to integrate xe^{-kx} twice then substitute correctly.

Areas for students to improve include:

- carefully developing the ΔV expression for cylindrical shells
- taking care with constants and negative signs when integrating e^{-kx}
- understanding that the mark on a question indicates the amount of work required.

Part (d)(i)

Students should:

- use the given graph and the properties of the absolute value function to produce the desired graph.

In better responses, students were able to:

- provide a sketch showing a cusp at $x = 1$ and a point of inflection at $y = -1$.

Areas for students to improve include:

- clearly showing key features of graphs such as cusps and points of inflexion.

Part (d)(ii)

Students should:

- recognise the features of the graph $y = \frac{1}{x^3 - 1}$.

In better responses, students were able to:

- clearly show the sketch of $y = \frac{1}{x^3-1}$ has an asymptote at $x = 1$ and a point of inflexion at $y = -1$ and approaches $y = 0$ as $x \rightarrow \pm\infty$.

Areas for students to improve include:

- clearly showing the key features graphs such as points of inflexions and asymptotes.

Part (d)(iii)

Students should:

- recognise the features of the graph $y = \frac{x}{x^3-1}$.

In better responses, students were able to:

- provide a sketch with asymptote at $x = 1$, curve approaching $y = 0^+$ as $x \rightarrow \pm\infty$ with left hand branch passing through $(0,0)$ and a maximum turning point in the 2nd quadrant.

Areas for students to improve include:

- clearly showing that the curve approaches the x axis asymptotically as $x \rightarrow -\infty$
- only having one graph on the set of axes.

Question 13

Part (a)(i)

Students should:

- be able to differentiate implicitly, substitute values, use the point gradient formula and rearrange into the required form using a Pythagorean Identity.

In better responses, students were able to:

- correctly rearrange the equation in a minimum numbers of steps, while still covering all important aspects.

Areas for students to improve include:

- rearranging equations involving trigonometric ratios and correctly using negative signs.

Part (a)(ii)

Students should:

- know how to solve simultaneous equations involving trigonometric ratios and other algebraic constants.

In better responses, students were able to:

- multiply the equations by $\tan \theta$ and $\tan \phi$ respectively, to eliminate the y , or make y the subject of both equations.

Areas for students to improve include:

- determining whether the elimination or substitution method will lead to an easier solution, again with reference to equations with trigonometric ratios and where fractions could be avoided for most of the solution if a more efficient method was chosen.

Part (a)(iii)

Students should:

- know how to calculate gradients and midpoints
- know different techniques to prove two expressions are equal.

In better responses, students were able to:

- choose OM and OT as the two pairs involving the simplest algebraic manipulation
- equate the two gradients by:
 - multiplying one of the gradients by the conjugate of either the numerator or denominator,
 - equating the two gradients then showing the left and right hand sides are equal,
 - proving that one gradient divided by the other equals one.

Areas for students to improve include:

- knowing, and correctly using, the midpoint and gradient formulae
- operating with equations involving tangent and secant ratios.

Part (b)(i)

Students should know:

- the formula for the area of a trapezium
- how to find an expression for an unknown side in terms of a given expression using similar triangles or basic trigonometry.

In better responses, students were able to:

- draw a diagram clearly showing the lengths of h , a and b
- write out the formula for the area of a trapezium, then substitute in the letters from the diagram, then the numbers from the diagram.

Areas for students to improve include:

- being aware that where the answer has been given, students need to clearly indicate what is being done and why, rather than writing down algebraic terms without context.

Part (b)(ii)

Students should:

- know how to calculate volumes by slices, including how to determine the limits and finding the correct primitive function.

In better responses, students were able to:

- find the correct integral and primitive function, then substitute correctly.

Areas for students to improve include:

- reviewing the diagram carefully to check limits, and taking more care when substitution involves negatives.

Part (c)(i)

Students should:

- know how to find the time of flight and range.

In better responses, students were able to:

- find the time of flight by letting $y = 0$, then substitute into $\dot{x} = 10t$.

Areas for students to improve include:

- multiplying with fractions involving surds.

Part (c)(ii)

Students should:

- be able to substitute constants from the previous part of a question into equations of motion to create simultaneous equations
- know how to correctly convert worded questions into symbols.

In better responses, students were able to:

- clearly state the time for Object 2, substitute into the equations of motion and rearrange to make $U \sin \alpha$ and $U \cos \alpha$ the subjects
- then divide the equations to find $\tan \alpha$, and substitute back in.

Areas for students to improve include:

- using different pronumerals for different objects
- understanding the difference between the constants and variables when substituting into equations.

Question 14

Part (a)

Students should:

- create balanced forces as $F = 0$ and resolve to the required equation
- drawn a diagram and analyse the forces.

In better responses, students were able to:

- balance horizontally and vertically or parallel to the track and resolve.

Areas for students to improve include:

- applying trigonometry correctly to the described forces
- using the correct notation and variables for the forces.

Part (b)(i)

Students should:

- use $\ddot{x} = 0$ at terminal velocity.

In better responses, students were able to:

- clearly demonstrate the solution to $g - kv = 0$.

Areas for students to improve include:

- understanding terminal velocity and its implications for $\ddot{x}$
- using the fact that $g - kv = 0$ rather than deriving an equation for v in terms of t and taking $\lim_{t \rightarrow \infty} e^{-kt} = 0$.

Part (b)(ii)

Students should:

- take the reciprocal of $\frac{dv}{dt}$ and integrate, either finding C or with limits, then substitution to solve.

In better responses, students were able to:

- correctly use limits
- recognise the need for an appropriate substitution
- successfully execute the appropriate integration technique and could simplify the expression using logarithm laws.

Areas for students to improve include:

- taking care with signs and with substituting limits as listed
- recognising the interrelatedness of part (i) and part (ii)
- attending to detail in order to avoid transcription errors
- using appropriate variables.

Part (b)(iii)

Students should:

- make an appropriate fraction split and integrate, or use the result from part (ii) to isolate v and integrate.

In better responses, students were able to:

- manipulate the expressions correctly and efficiently to achieve accurate integrals, substitutions, leading to the correct result

- clearly set out, showing all steps of their working.

Areas for students to improve include:

- recognising the appropriate integration technique such as $v \frac{dv}{dx} = f(v)$
- simplifying algebra using techniques such as

$$-\frac{1}{k} \int \frac{g}{g-kv} dv = \frac{1}{k} \int \frac{g}{kv-g} dv \text{ or } -\frac{1}{k} \int_a^b \frac{g}{g-kv} dv = \frac{1}{k} \int_a^b \frac{g}{kv-g} dv$$
- taking care with signs
- taking care with substituting limits as listed
- recognising the interrelatedness of parts ii and iii in a question.

Part (c)(i)

Students should:

- substitute for double angles and manipulate the equation toward the required conclusion
- learn the correct substitutions or make use of the Reference Sheet.

In better responses, students were able to:

- know their trigonometric identities, including t -formulae
- use LHS and RHS in their proof
- know how to manipulate fractions, brackets and minus signs.

Areas for students to improve include:

- working with negative signs
- substituting a trigonometric identity correctly
- being efficient in their use of identities
- working from LHS to RHS or vice versa to avoid final statements such as $\sin x = \sin x$.

Part (c)(ii)

Students should:

- understand the process of induction and manipulate the LHS to match the RHS or vice versa.

In better responses, students were able to:

- recognise the double angle and use either part (i) or trigonometric identities to solve
- make reference to the identity used from part (i) and show the reasoning
- manipulate the index laws correctly to arrive at the correct conclusion.

Areas for students to improve include:

- recognising the interrelatedness of parts (i) and (ii) in a question
- understanding that a series is part of the statement required to be proved rather than single terms

- being able to generalise to identify how the use of part (i) could be used in a different context.

Question 15

Part (a) (i)

Students should:

- be careful to change limits of integrals when making a substitution
- clearly show all steps in a proof.

In better responses, students were able to:

- use the substitution $u = -x$ and substitute correctly clearly showing the substitution step.

Areas for students to improve include:

- knowing when a substitution is more appropriate to use rather than odd or even functions when looking at $f(x) = f(-x)$
- reading the question carefully to know what they are trying to prove
- realising that dx and du are a critical part of an integral
- understanding it is not sufficient to prove something for an even function if you do not know what type of function you have.

Part (a) (ii)

Students should:

- understand how part (i) relates to part (ii)
- look for the most efficient method to gain an answer.

In better responses, students were able to:

- use part (i) to gain $\int_{-1}^1 \frac{e^x}{e^x + e^{-x}} dx = \int_{-1}^1 \frac{e^{-x}}{e^{-x} + e^x} dx$, then solve to gain $I = 1$.

Areas for students to improve include:

- recognising that using part (i) would produce the most efficient method to solve the integral.

Part (b) (i)

Students should:

- draw a diagram when approaching a probability problem.

In better responses, students were able to:

- recognise that the game may continue past the first two turns
- understand the significance of A having a probability of $\frac{1}{2}$ of winning or losing on the first turn

- recognise the pattern and that the probabilities were added $P(A \text{ winning}) = \frac{1}{2} + \frac{1}{8} + \dots$.

Areas for students to improve include

- reading questions carefully to determine whether selection is with or without replacement
- understanding what is occurring long term, in a probability question by drawing a diagram
- writing clear and concise answers, especially when question is only worth 1 mark.

Part (b) (ii)

Students should:

- use correct probability notation
- interpret the problem with a diagram and recognize what is occurring long term.

In better responses, students were able to:

- write out a number of draws in the game
- consider $P(B \text{ winning}) > \frac{1}{2}$ rather than $P(B) > P(A)$
- recognise that the question involved a limiting sum not a sum to n terms then solve the inequality $y^2 - wy - w^2 > 0$ in terms of $\frac{y}{w}$.

Areas for students to improve include:

- interpreting the language and meaning of probability questions
- recognising a limiting sum of a GP and using appropriate formula
- taking care when simplifying fractional algebraic expressions
- recognising that the question involved a limiting sum not a sum to n terms.

Part (c) (i)

Students should:

- know the most efficient method to solve an integral of the form $\frac{Ax}{(x+1)^2}$.

In better responses, students were able to:

- use integration by parts, write the integral $\int_0^1 \frac{x}{(x+1)^2} dx = \int_0^1 \frac{x+1-1}{(x+1)^2} dx$, or substitute $x = u + 1$ to gain $\ln 2 = -\frac{1}{2}$.

Areas for students to improve include:

- taking care when substituting the limits of integration into a fractional expression
- ensuring that the answer obtained matches the answer given in a 'show' question
- taking care with negative signs and removing fractions from brackets.

Part (c) (ii)

Students should:

- recognise the need to use integration by parts to solve this recurrence relation
- know techniques to split an expression into the parts that are required.

In better responses, students were able to:

- use integration by parts to gain an expression in x^{n-1} then split $n \int_0^1 \frac{x^{n-1}}{x+1} dx$ into $n \int_0^1 \frac{x^n}{(x+1)^2} + \frac{x^{n-1}}{(x+1)^2} dx$ then rearrange to gain the correct answer.

Areas for students to improve include:

- using the correct choices for u and $\frac{dv}{dx}$ in integration by parts
- taking care with negative signs in particular $(1 - n) = -(n - 1)$
- making sure the final answer obtained matches the answer given
- recognising the best technique to gain I_n and I_{n-1} from their integral expression.

Part (c) (iii)

Students should:

- recognise the significance of parts i and ii to help solve this recurrence relation rather than starting again.

In better responses, students were able to:

- use $I_2 = \frac{1}{2} - 2(\ln 2 - \frac{1}{2})$ combining parts (i) and (ii) then find I_3 .

Areas for students to improve include:

- recognising the relevance of part (i) and (ii) and use them rather than starting again
- taking care with calculations involving order of operations and fractions
- using the answer given from part (ii), not their answer.

Question 16

Part (a) (i)

Students should:

- be able to substitute given results into an expression and manipulate the resulting expression to achieve the desired result.

In better responses, students were able to:

- show, via direct substitution, that $x = r \cos \theta$ satisfies the equation $x^3 - px + q = 0$, making use of the given result, $4 \cos^3 \theta - 3 \cos \theta = \cos 3\theta$ (and the assumptions given in the question for r and $\cos 3\theta$)
- commence their proof with the given result, $4 \cos^3 \theta - 3 \cos \theta = \cos 3\theta$, and then proceed

to obtain the equation $(r \cos \theta)^3 - p(r \cos \theta) + q = 0$, thus showing that $r \cos \theta$ is a root of the equation $x^3 - px + q = 0$.

Areas for students to improve include:

- guarding against making the mistake of assuming true, the statement which is required to be proven
- showing all necessary working in their solution
- justifying any substitution of variables.

Part (a) (ii)

Students should:

- be able to use expand a cubic expression
- use the relationship between roots and coefficients of a polynomial equation.

In better responses, students were able to:

- show, by expansion, that the equation $(x - 3)^3 + 9(x - 3)^2 + 15(x - 3) - 17 = 0$ (whose roots are $\alpha + 3, \beta + 3$ and $\gamma + 3$) is equivalent to the equation $x^3 - 12x - 8 = 0$
- show, by expansion, that the equation $(\alpha + 3)^3 - 12(\alpha + 3) - 8 = 0$ is equivalent to $\alpha^3 + 9\alpha^2 + 15\alpha - 17 = 0$ to reach the appropriate conclusion
- find the sum, sum of products and product of the roots $\alpha + 3, \beta + 3$ and $\gamma + 3$.

Areas for students to improve include:

- providing all necessary algebraic working such as expanding cubic expressions in order to prove a given result
- finding all three items (sum, sum of products and product of roots) to establish the appropriate cubic equation.

Part (a) (iii)

Students should:

- be able to solve a given cubic by making use of trigonometric substitution and algebraic manipulation.

In better responses, students were able to:

- use the results of both part (i) and part (ii), to solve the equation
- first solve the equation $x^3 - 12x - 8 = 0$ by putting $p = 12, q = -8$ in part (i), followed by finding the values of r and θ
- use part (ii) to write down the solution in terms of the roots of the equation.
 $x^3 - 12x - 8 = 0$.

Areas for students to improve include:

- finding all possible roots in a guided question involving cubic equations.

Part (b) (i)

Students should:

- be able to relate the complex roots of a given polynomial to the coefficients of the polynomial.

In better responses, students were able to:

- find the sum, $\operatorname{Re}(\alpha) + \operatorname{Re}(\beta)$, from the sum of roots of $P(z) = 0$
- find the product, $\operatorname{Re}(\alpha) \times \operatorname{Re}(\beta)$, from the sum of the products of pairs of roots of $P(z) = 0$
- find both the sum and the product of $\operatorname{Re}(\alpha)$ and $\operatorname{Re}(\beta)$ in order to find the sum of their squares
- find $(\operatorname{Re}(\alpha))^2 + (\operatorname{Re}(\beta))^2$ by expanding and finding $\left(\frac{\alpha + \bar{\alpha}}{2}\right)^2 + \left(\frac{\beta + \bar{\beta}}{2}\right)^2$.

Areas for students to improve include:

- knowing and applying basic complex number identities such as $\operatorname{Re}(\alpha) = \frac{1}{2}(\alpha + \bar{\alpha})$ and $\alpha\bar{\alpha} = |\alpha|^2$.

Part (b) (ii)

Students should:

- be able to determine the four possible positions of β and how they relate to the position of α .

In better responses, students were able to:

- show all four possible positions of β in relation to the position of α , with reference to either $\arg \alpha$ or the coordinates of α : $(\operatorname{Re}(\alpha), \operatorname{Im}(\alpha))$.

Areas for students to improve include:

- responding appropriately to key directions in an examination question such as 'accurately show'.

Part (c)

Students should:

- be able to use appropriate circle geometry theorems and trigonometric results to establish the given result.

In better responses, students were able to:

- apply the sine rule in two different triangles such as $\triangle ABC$ and $\triangle BCD$ to establish a key result $\frac{a}{\sin DBC} = \frac{d}{\sin BCA}$ (or something similar) using the geometric property of equal angles in the same segment of a circle
- establish, through geometrical reasoning, applicable results involving angles $B + E$ and $C + E$ such as $\angle DBC = B + E - \pi$ and $\angle BCA = C + E - \pi$.

Areas for students to improve include:

- making use of appropriate circle geometry theorems and properties
- stating the reasoning used in making geometrical assertions.